

CANADIAN OLYMPIC PARK UPPER, AB, Canada

WMO: 712350

Lat: 51.076N

Lon: 114.212W

Elev: 4052

StdP: 12.67

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.9	-11.4	-23.0	1.8	-16.6	-17.1	2.5	-10.6	25.3	33.5	22.5	33.6	5.6	280	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	82.9	59.5	79.3	58.4	75.9	57.4	63.3	75.7	61.2	74.1	59.4	71.8	7.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.0	86.5	68.0	56.5	79.1	65.7	54.4	73.3	64.1	30.9	75.6	29.3	74.4	27.9	72.2	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.9	18.9	16.5		DB	-21.6	89.3	6.3	3.3	-26.1	91.7	-29.8	93.7	-33.3	95.5	-37.9	98.0
				WB	-22.0	67.2	6.0	4.5	-26.3	70.4	-29.8	73.1	-33.1	75.6	-37.4	78.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.6	21.2	22.1	29.2	39.4	49.1	55.7	62.1	60.7	52.5	41.5	29.6	22.2
	DBStd	18.52	16.14	14.56	14.06	9.78	8.77	6.31	5.98	6.57	8.99	9.56	13.09	14.08
	HDD50	4647	892	781	646	336	123	18	2	6	76	290	614	863
	HDD65	9006	1357	1201	1110	768	494	286	127	163	381	729	1062	1328
	CDD50	1198	0	0	1	18	95	189	377	337	152	27	2	0
	CDD65	83	0	0	0	0	2	7	38	29	7	0	0	0
	CDH74	1121	0	0	0	4	48	90	476	382	114	7	0	0
	CDH80	245	0	0	0	0	4	15	117	86	20	3	0	0
(14) Wind	WSAvg	7.6	7.8	7.4	7.7	8.1	8.0	7.7	7.2	6.7	7.0	7.7	7.6	7.8
(15) Precipitation	PrecAvg	18.0	0.5	0.4	0.7	1.2	2.8	4.0	2.6	2.3	1.7	0.7	0.6	0.6
	PrecMax	24.6	1.0	0.9	2.0	3.1	4.9	11.9	6.1	4.2	4.1	1.7	1.7	1.5
	PrecMin	12.1	0.1	0.0	0.2	0.3	1.1	1.3	0.8	0.4	0.3	0.1	0.0	0.1
	PrecStd	2.6	0.2	0.2	0.4	0.7	1.2	2.0	1.4	1.0	0.9	0.4	0.4	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	53.5	60.8	71.0	79.3	82.3	88.3	87.1	82.9	71.1	59.6	51.9
		MCWB	40.1	39.7	43.5	48.4	54.1	59.9	60.6	59.2	56.2	50.7	44.1	38.1
	2%	DB	47.7	48.2	55.0	64.8	74.1	76.2	83.2	82.3	77.2	64.5	53.4	46.8
		MCWB	37.0	36.8	40.6	45.6	52.2	57.0	60.8	59.1	55.1	48.1	41.3	36.3
	5%	DB	44.4	44.0	50.7	59.8	69.2	72.1	79.4	78.2	72.4	59.2	49.5	43.4
		MCWB	35.2	34.3	38.3	43.3	50.0	55.1	60.2	58.3	53.6	45.1	38.9	34.2
	10%	DB	41.2	40.2	46.5	54.8	64.9	68.7	75.5	74.4	67.8	55.3	45.5	39.9
		MCWB	33.1	31.7	36.0	40.5	48.2	53.5	59.0	57.5	51.7	43.0	36.2	31.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.4	40.6	44.5	49.6	56.0	62.7	66.5	65.6	59.3	52.5	45.7	40.0
		MCDB	51.0	51.5	56.9	66.9	73.0	77.2	77.4	76.0	75.5	68.4	56.9	49.5
	2%	WB	37.7	37.2	41.5	46.5	53.6	59.4	64.0	62.5	56.8	49.1	42.0	37.0
		MCDB	46.5	46.9	53.9	62.5	70.1	71.9	76.1	74.6	72.6	62.2	52.3	45.9
	5%	WB	35.5	34.7	38.8	44.0	51.5	57.2	62.1	60.5	54.8	46.2	39.3	34.4
		MCDB	43.7	43.6	49.6	58.2	66.0	68.6	75.1	73.5	69.5	57.7	48.8	42.9
	10%	WB	33.2	31.9	36.4	41.4	49.5	55.0	60.3	58.7	52.6	43.8	36.6	32.0
		MCDB	41.0	39.6	46.0	53.2	62.8	65.8	72.7	71.4	65.7	54.0	44.8	39.5
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	17.4	18.1	19.1	20.4	19.4	21.9	21.9	20.6	18.0	15.7	15.2
		MCDBR	18.3	21.0	23.4	26.6	27.4	25.0	27.6	28.2	28.1	23.8	21.2	18.2
	5% WB	MCWBR	12.2	13.2	13.5	13.8	12.8	12.0	12.2	12.3	13.0	12.9	13.0	12.1
		MCDWR	17.9	21.0	23.0	25.4	24.6	22.6	24.3	24.2	25.8	23.2	21.2	18.2
(40) Clear-Sky Solar Irradiance	taub		0.246	0.265	0.275	0.329	0.336	0.340	0.335	0.350	0.306	0.274	0.249	0.233
		taud	2.439	2.449	2.489	2.338	2.387	2.412	2.427	2.397	2.491	2.564	2.511	2.425
	Ebn at Noon		258	282	299	289	292	290	290	279	283	276	257	246
		Edh at Noon	18	23	27	36	36	35	34	33	27	21	16	15
(44) All-Sky Solar Radiation	RadAvg		357	664	1059	1464	1721	1830	2030	1669	1177	727	387	279
	RadStd		24	49	75	104	136	139	132	117	129	75	28	20

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (65 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page